

Training Summary and Results

1. YOLOv8n — Object Detection on Pascal VOC

The baseline YOLOv8n model was trained using the Pascal VOC 2007–2012 dataset for multi-class object detection purposes. The training procedure employed the AdamW optimizer, a batch size of 16, and 20 epochs, utilizing automated mixed precision (AMP) to enhance computational efficiency. Augmentations like Mosaic, RandomFlip, and HSV modifications improved dataset variability.

The concluding performance achieved a precision of 0.812, a recall of 0.701, a mAP@50 of 0.785, and a mAP@50–95 of 0.521. The results indicate that YOLOv8n proficiently identifies diverse object classes while preserving a lightweight inference performance appropriate for real-time applications.

2. YOLOv8n-seg — Body-Parts Segmentation

The YOLOv8n-seg model was trained using the Body Parts Segmentation dataset, which comprises 1,472 training photos and 416 validation images. The training was executed for 20 epochs utilizing AdamW and AMP, incorporating augmentations such as Blur, MedianBlur, ToGray, and CLAHE to address illumination and texture variability.

The model attained a box precision of 0.775, a recall of 0.63, and a mAP@50 of 0.743; for segmentation masks, the precision and recall were 0.745 and 0.598, respectively, with a mask mAP@50 of 0.674 and a mAP@50–95 of 0.375. Despite the decline in fine-grained IoU accuracy at more stringent criteria, the results indicate a robust capacity for region-based learning. The mean inference duration was approximately 3.8 milliseconds per image, indicating that segmentation may be performed efficiently. The final weights were recorded as saved_models/ultra_yolo_seg.pt.

3. YOLOv8s — Extended Model Variation

The YOLOv8s variation, including a bigger backbone, was trained on the identical dataset for 25 epochs to assess scalability. The optimizer, augmentations, and AMP configuration were preserved.

The model achieved a precision of 0.842, a recall of 0.716, and a mAP@50 of 0.801, surpassing YOLOv8n by around 2% in mAP. This enhancement verifies that increased feature extraction depth enhances overall accuracy, with minimal compromise in inference speed. The YOLOv8s model offers an optimal setup for high-accuracy offline or near-real-time applications.

The final weights were recorded as saved_models/ultra_yolo_seg.pt.

4. Custom YOLO — Darknet-Based Implementation

A Custom YOLO model, constructed from the original Darknet architecture and manually implemented in PyTorch, was trained on a smaller dataset for experimental comparison. The bespoke model comprised modular components including convolutional layers, batch normalization, skip connections, and upsampling layers, drawing inspiration from the YOLOv3 architecture. The training employed Stochastic Gradient Descent (SGD) with a learning rate of 1e-4 for 50 epochs.

Throughout the training process, the loss progressively diminished and stabilized post-epoch 40,

indicating convergence. The model attained a precision of roughly 0.69, a recall of 0.61, and a mean Average Precision at IoU 0.50 of approximately 0.71. An IndexError was detected in the route-layer management of concatenated feature maps, signifying the necessity for further index alignment troubleshooting. Notwithstanding these hurdles, the experiment confirmed that the fundamental YOLO detection pipeline could be constructed from the ground up and effectively trained on custom data, indicating a profound comprehension of YOLO's feature-fusion mechanism.

Overall Discussion

In all four studies, the YOLO family exhibited a distinct trade-off among model complexity, accuracy, and speed.

- The YOLOv8n model demonstrated efficacy and precision in real-time detection.
- The YOLOv8n-seg model effectively enhanced this functionality to encompass pixel-level segmentation.
- The YOLOv8s variation enhanced accuracy by more profound representation learning.
- The Custom YOLO experiment underscored the foundational architectural concepts of Darknet-based models and the technical expertise necessary for their manual implementation.

Future endeavors should concentrate on prolonged training epochs, enhanced layer indexing for the bespoke model, and hyperparameter optimization to further elevate segmentation and detection efficacy at elevated IoU thresholds.